

Norway

SLEIPNER

Volve F

F-5

F-5

Plan: F5 Rev 14.3 131207

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table (As Installed) @ 54.90m
Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m
Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F5 Rev 14.3 131207	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-5		
Well Position	+N/-S	-4.65 m	Northing:
	+E/-W	0.92 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.6569 N
			Longitude:
			1° 53' 14.9902 E
			Water Depth:
			91.00 m

Wellbore	F-5		
Magnetics	Model Name	Sample Date	Declination
			(°)
			Dip Angle
			(°)
			Field Strength
			(nT)
	MNETICREFERENCE	01.06.2008	-2.14
			71.62
			50,352

Design	F5 Rev 14.3 131207		
Audit Notes:			
Version:	Phase:	PLAN	Tie On Depth:
			145.90
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	-4.65	0.92
			Direction
			(°)
			81.75

Survey Tool Program	Date	31.05.2010		
From	To	Survey (Wellbore)	Tool Name	Description
(m)	(m)			
145.90	1,415.00	F5 Rev 14.3 131207 (F-5)	Keeper, stat	Gyro Tool from SDC
1,515.00	3,868.30	F5 Rev 14.3 131207 (F-5)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Planned Survey									
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)
145.90	0.00	0.00	145.90	-4.65	0.92	0.00	0.000	0.000	0.000
220.00	0.00	0.00	220.00	-4.65	0.92	0.00	0.000	0.000	0.000
370.00	5.00	190.00	369.81	-11.10	-0.21	-2.05	1.000	1.000	0.000
485.38	10.00	190.00	484.17	-25.92	-2.83	-6.76	1.300	1.300	0.000
899.80	25.00	115.00	882.00	-99.47	71.41	56.16	1.758	1.086	-5.429
1,112.76	25.00	115.00	1,075.00	-137.51	152.98	131.43	0.000	0.000	0.000
1,287.72	31.65	102.93	1,229.06	-163.45	231.37	205.29	1.500	1.140	-2.070
1,398.81	32.51	109.72	1,323.21	-180.05	287.90	258.85	1.000	0.232	1.833
2,178.46	32.51	109.72	1,980.70	-321.40	682.33	628.94	0.000	0.000	0.000
2,289.54	31.65	102.93	2,074.86	-338.00	738.86	682.50	1.000	-0.232	-1.833

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Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F5 Rev 14.3 131207	Database:	Production EDM P246N

Planned Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,541.89	31.04	86.76	2,290.76	-349.16	868.57	809.27	1.000	-0.073	-1.922
2,760.97	31.04	86.76	2,478.48	-342.78	981.34	921.79	0.000	0.000	0.000
2,937.15	30.52	74.16	2,630.00	-327.99	1,069.82	1,011.47	1.100	-0.088	-2.146
3,554.54	53.04	19.54	3,102.01	-40.26	1,313.72	1,294.11	2.000	1.094	-2.654
3,592.02	55.00	17.62	3,124.03	-11.51	1,323.37	1,307.79	2.002	1.569	-1.537
3,637.62	51.96	17.62	3,151.16	23.41	1,334.46	1,323.78	2.000	-2.000	0.000
3,786.67	51.97	17.61	3,243.00	135.31	1,370.00	1,374.99	0.002	0.002	-0.001
3,868.30	51.97	17.61	3,293.29	196.59	1,389.45	1,403.04	0.000	0.000	0.000

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Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
F5 Tgt Rev 2 101106 - plan misses target center by 125.63m at 3300.96m MD (2928.68 TVD, -204.28 N, 1230.71 E) - Point	0.00	0.00	3,004.90	-263.61	1,150.38	6,478,300.00	436,200.00	58° 26' 21.8979 N	1° 54' 26.0898 E
F5 Pilot - plan misses target center by 978.04m at 3396.62m MD (2998.61 TVD, -149.09 N, 1265.43 E) - Point	0.00	0.00	3,064.90	-843.82	1,950.66	6,477,720.00	437,000.00	58° 26' 3.5664 N	1° 55' 15.9815 E
F5 Top Hugin RTD - plan misses target center by 0.77m at 3554.54m MD (3102.01 TVD, -40.26 N, 1313.72 E) - Point	0.00	0.00	3,102.00	-40.54	1,314.44	6,478,523.00	436,364.00	58° 26' 29.1926 N	1° 54' 35.9771 E
F5 Hugin T1 230907 - plan misses target center by 67.95m at 3511.12m MD (3075.23 TVD, -72.23 N, 1301.64 E) - Point	0.00	0.00	3,104.55	-132.57	1,312.44	6,478,431.00	436,362.00	58° 26' 26.2176 N	1° 54' 35.9457 E
F5 Top Target 291007 - plan misses target center by 0.71m at 3592.54m MD (3124.33 TVD, -11.10 N, 1323.50 E) - Point	0.00	0.00	3,124.90	-11.53	1,323.44	6,478,552.00	436,373.00	58° 26' 30.1348 N	1° 54' 36.5030 E
Volve I30 Topp - plan misses target center by 154.67m at 3554.54m MD (3102.01 TVD, -40.26 N, 1313.72 E) - Point	0.00	0.93	3,133.83	-121.57	1,441.38	6,478,442.00	436,490.90	58° 26' 26.6407 N	1° 54' 43.8814 E
F5 040107 - plan misses target center by 41.15m at 3663.76m MD (3167.27 TVD, 43.03 N, 1340.70 E) - Point	0.00	0.00	3,134.90	66.50	1,350.45	6,478,630.00	436,400.00	58° 26' 32.6704 N	1° 54' 38.0896 E
F5 1809 Pt 3 - plan misses target center by 201.83m at 3549.97m MD (3099.25 TVD, -43.70 N, 1312.49 E) - Point	0.00	0.00	3,192.70	-220.50	1,339.75	6,478,343.10	436,389.30	58° 26' 23.3905 N	1° 54' 37.7166 E
F5 1809 Pt 2 - plan misses target center by 143.73m at 3590.25m MD (3123.01 TVD, -12.89 N, 1322.93 E) - Point	0.00	0.00	3,217.80	-104.66	1,379.96	6,478,458.90	436,429.50	58° 26' 27.1549 N	1° 54' 40.0792 E
Volve I31 Bunn - plan misses target center by 198.07m at 3605.06m MD (3131.59 TVD, -1.39 N, 1326.59 E) - Point	0.00	0.93	3,239.34	-121.57	1,441.38	6,478,442.00	436,490.90	58° 26' 26.6407 N	1° 54' 43.8814 E
F5 1809 Pt 1 - plan misses target center by 68.00m at 3718.49m MD (3200.99 TVD, 84.11 N, 1353.74 E) - Point	0.00	0.00	3,240.10	40.89	1,388.76	6,478,604.40	436,438.30	58° 26' 31.8629 N	1° 54' 40.4765 E
F5 Base Hugin RTD - plan hits target center - Rectangle (sides W47.00 H65.00 D0.00)	25.00	65.00	3,243.00	135.31	1,370.00	6,478,698.79	436,419.54	58° 26' 34.9040 N	1° 54' 39.2256 E
F5 Target Base 291007 - plan misses target center by 2.30m at 3786.67m MD (3243.00 TVD, 135.31 N, 1370.00 E) - Point	0.00	0.00	3,244.90	136.52	1,370.46	6,478,700.00	436,420.00	58° 26' 34.9437 N	1° 54' 39.2528 E
F5 Bae Hugin T2 230907 - plan misses target center by 10.79m at 3786.67m MD (3243.00 TVD, 135.31 N, 1370.00 E) - Point	0.00	0.00	3,251.00	142.53	1,370.46	6,478,706.00	436,420.00	58° 26' 35.1376 N	1° 54' 39.2468 E

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Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F5 Rev 14.3 131207	Database:	Production EDM P246N

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.80	220.80	30"	30.000	36.000	
1,404.70	1,328.18	13 3/8"	13.375	17.500	
2,921.00	2,616.09	9 5/8"	9.625	12.250	
3,790.10	3,245.11	7"	7.000	8.500	

Formations						
Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)	
894.29	877.00	Utsira Top		0.00		
1,098.41	1,062.00	Utsira Base		0.00		
1,260.79	1,206.00	Skade Top		0.00		
1,380.79	1,308.00	Skade Base		0.00		
2,297.93	2,082.00	Grid Top		0.00		
2,371.74	2,145.00	Grid Base		0.00		
2,515.33	2,268.00	Balder Top		0.00		
2,605.19	2,345.00	Sele Top		0.00		
2,682.22	2,411.00	Lista Top		0.00		
2,753.41	2,472.00	Ty Top		0.00		
2,916.25	2,612.00	Ekofisk Top		0.00		
2,931.34	2,625.00	Tor Top		0.00		
3,080.42	2,752.00	Hod Top		0.00		
3,177.54	2,832.00	Blodoeks Top		0.00		
3,208.70	2,857.00	Hidra Top		0.00		
3,220.03	2,866.00	Roedby Top		0.00		
3,253.15	2,892.00	Aasgard Top		0.00		
3,292.15	2,922.00	Draupne Top		0.00		
3,493.51	3,064.00	Heather Top		0.00		
3,554.53	3,102.00	Hugin Top		0.00		
3,786.67	3,243.00	Sleipner Top		0.00		

Checked By: _____ Approved By: _____ Date: _____